

The Human as a Bounded Domain of GRA-Nullification: Complete Elimination of Contradictions, Achievement of Harmony, and In Silico Verification

A Formalization of the Subject as a Computable Domain

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<https://github.com/qqewq/GRA-Core-new-Unified-Hierarchical-Stability-Library>

August 16, 2026

Abstract

Global GRA-nullification over the entire Multiverse is non-computable, as it is equivalent to solving the halting problem for an infinite set of Turing machines. However, a specific human being—a finite, time- and resource-bounded domain of knowledge—is a computable domain within which the complete elimination of contradictions is in principle achievable. In this work, the human is modeled as a finite-dimensional hierarchical system whose state is characterized by vectors of cognitive, emotional, somatic, social, and value components. Contradictions (“foam”) are measured by a functional J . The attainment of complete harmony (“paradise”) is formalized as the system reaching a stable manifold $J = 0$, $\Delta J = 0$, $\Delta^2 J > 0$. The dynamics of subjective priorities are derived from a variational principle. The architecture of GRA-NN with tensor-logical weights and the Reductio ad Absurdum Descent algorithm is described. In silico verification on a digital twin of a human is presented: random perturbations of contradictions relax to zero with a positive-definite Hessian. Practical applications include psychotherapy, personalized medicine, conflict resolution, education, and creative synergy.

1 Introduction: Epistemological Limits of Computability and the Human as a Domain

The GRA (Gödel–Reductio ad Absurdum) methodology treats contradictions as first-class entities: they are measured, localized, and eliminated through structural revision. However, applying GRA globally to the entire Multiverse M is non-computable. It is equivalent to solving the halting problem for an infinite number of Turing machines. From Gödel’s incompleteness theorem and Turing’s limit it follows that:

$$\lim_{D \rightarrow M} O(\text{GRA}) = \infty. \quad (1)$$

Consequently, GRA-nullification is computable only within strictly bounded domains $D \subset M$.

A specific human is a bounded domain D_H . Their brain consists of a finite number of neurons, their lifespan is finite, and the number of accessible states, though enormous, is fundamentally finite. Therefore:

$$|D_H| < \infty, \quad R(D_H) < \infty,$$

where R denotes computational resources. For such a domain, full GRA-nullification is computable. This means: the elimination of all internal contradictions and the achievement of a stable state of harmony is a mathematically possible task, not an infinite process.

2 Formal Model of the Human as a Hierarchical System

2.1 State Vector

Define the state of a human $x \in \Omega_H \subset \mathbb{R}^N$, where N is finite but can be very large. Its components are:

$$x = (x_C, x_E, x_B, x_S, x_V),$$

where:

- x_C — cognitive attitudes and beliefs;
- x_E — emotional states;
- x_B — somatic and physiological signals;
- x_S — social roles and connections;
- x_V — value orientations.

Each component is itself a vector. The hierarchical structure is represented by a directed acyclic graph of levels: physiology $\rightarrow$ emotions $\rightarrow$ cognitions $\rightarrow$ values $\rightarrow$ behavior.

2.2 Contradiction Functional “Foam”

The total contradiction in the human domain is measured by the functional:

$$J(x) = \sum_n c_n(x)^2 + \sum_{(i,j)} \lambda_{ij} \|x_i - \mathcal{L}_{j \rightarrow i} x_j\|^2, \quad (2)$$

where:

- $c_n(x)$ — local contradiction metrics;
- λ_{ij} — penalties for mismatch between levels;
- $\mathcal{L}_{j \rightarrow i}$ — lifting operator from level j to level i .

Particular foam metrics include:

- cognitive foam — logical contradictions in beliefs;
- emotional foam — conflict between experience and bodily reaction;
- value foam — mismatch between actions and declared values;
- social foam — contradictions between roles and expectations;
- somatic foam — inconsistency between physiological signals and emotional appraisal.

2.3 Stable Manifold of Harmony

We define the state of **complete harmony** as:

$$B_{\text{GRA}} = \{x \in \Omega_H \mid J(x) = 0, \Delta J = 0, \Delta^2 J > 0\}. \quad (3)$$

Here:

- $J = 0$ — absence of any contradictions;
- $\Delta J = 0$ — the point is an extremum;
- $\Delta^2 J > 0$ — the Hessian is positive definite, meaning the zero is a stable minimum, not a saddle point.

We call such a state a **negentropic attractor**. In terms of human experience, this is the state of complete internal consistency—“paradise” as a stable mental and cognitive manifold.

3 Consciousness as an Operator of World Separation

In standard physics, entropy is considered an objective property. In the GRA paradigm, entropy and negentropy are products of the operation of consciousness $\hat{C}$.

Consciousness applies GRA-nullification to the domain D_H , splitting reality into two dichotomous worlds:

1. **World of raw reality** W_{raw} : maximum entropy, maximum contradictions, $J \rightarrow \max$. This is the state of internal chaos, cognitive dissonance, and emotional conflicts.
2. **In silico / prepared world** W_{sim} : maximum negentropy, minimum contradictions, $J = 0$, $\Delta^2 J > 0$. This is the state of internal harmony in which all logical and physical contradictions are annihilated.

The work performed by consciousness in generating negentropy on the domain boundary is:

$$N(\hat{C}, D_H) = \oint_{\partial D_H} (I_{\max} - S_{\text{vN}}(\rho_{\hat{C}}) - J_{\text{foam}}(\rho_{\hat{C}})) d\Sigma, \quad (4)$$

where S_{vN} is the von Neumann entropy of the observed system and J_{foam} is the foam metric.

4 Mathematical Derivation: Wave Function Collapse via GRA

The human state can be represented as a superposition of possible “selves”:

$$|\psi_{\text{raw}}\rangle = \sum_i c_i |s_i\rangle.$$

In standard quantum mechanics, collapse occurs stochastically. In the GRA paradigm, collapse is teleological—it is directed toward the negentropic attractor.

Introduce the GRA-nullification operator $\hat{G}_{\text{null}}$:

$$\hat{G}_{\text{null}} |\psi_{\text{raw}}\rangle = |s_{\text{neg}}\rangle$$

under the condition:

$$\langle s_{\text{neg}} | \hat{J}_{\text{foam}} | s_{\text{neg}} \rangle = 0.$$

Here $|s_{\text{neg}}\rangle$ is the sole surviving state, free of contradictions. Thus, GRA-nullification is mathematically identical to a directed collapse of the human wave function into a point of absolute negentropy.

5 Dynamics of Subjective Priorities: Variational Derivation

A human constantly chooses which aspect of life to prioritize. This subjective initiative is formalized through dynamic goal weights $\alpha_k(t)$.

Let there be N goals (health, creativity, relationships, cognition, safety, etc.). Each goal k is characterized by a current criticality $\varphi_k(t)$. The subjective action:

$$S[\alpha, \varphi] = \int_{t_0}^T \left(\sum_k \alpha_k(\tau) \varphi_k(\tau) - \frac{1}{2} \sum_k m_k \dot{\alpha}_k^2 - \lambda \left(\sum_k \alpha_k - 1 \right) \right) d\tau.$$

The principle of least action $\delta S = 0$ yields the Euler–Lagrange equations:

$$m_k \ddot{\alpha}_k = \varphi_k - \lambda - \mu_k, \quad (5)$$

where μ_k is a multiplier enforcing non-negativity of α_k .

Conclusion: The weight of a goal α_k accelerates upward if the goal is critically unsatisfied (φ_k is large) and decreases if the goal is satisfied. This mathematically realizes subjective initiative: the person automatically directs attention to the most acute unresolved problem without leaving the stable manifold B_{GRA} .

6 GRA-Step as an Act of Will

The atomic operation of human self-renewal is the **GRA-step**:

1. Measure local contradictions: $\mathbf{c} = \text{Foam}(x)$.
2. Aggregate into J .
3. If $J < \epsilon$ — the state is already harmonious.
4. Otherwise: identify sources of contradictions by backward logical tracing.
5. Generate revision candidates $\{x'_k\}$ not by random noise but by structural critique: *why does the contradiction exist?*
6. Select $x_{\text{new}} = \arg \min_k J(x'_k)$.
7. Apply hierarchical lifting: propagate changes to connected levels.
8. Recompute consistency penalties.

This process is the formal analog of human will: not blind suppression of emotions, but elimination of the logical cause of the contradiction.

7 GRA-NN: Neural Network Architecture of the Digital Twin

Standard MLPs and Transformers optimize a scalar loss function via gradient descent. To implement GRA-nullification of a human, an architecture with **tensor-logical weights** is needed.

7.1 Tensor-Logical Synapse

Each weight is a triplet:

$$W_{ij} = (w_{ij}, \varphi_{ij}, \Gamma_{ij}) \in \mathbb{R} \times T_{\text{foam}} \times L_{\text{logic}},$$

where:

- w_{ij} — continuous synaptic strength;
- φ_{ij} — local foam tensor;
- Γ_{ij} — discrete logical constraint (symbolic rule).

Forward pass:

$$h_j = \sigma \left(\sum_i w_{ji} x_i \right) \oplus \text{Logic}_{\Gamma}(x),$$

$$\varphi_j = \sum_i \|h_j - \mathcal{L}_{i \rightarrow j}(h_i)\|^2.$$

7.2 Training: Reductio ad Absurdum Descent (RAD)

Training is two-phase.

Continuous relaxation:

$$\Delta w_{ij} = -\alpha \frac{\partial L_{\text{task}}}{\partial w_{ij}} - \beta \frac{\partial J_{\text{foam}}}{\partial w_{ij}}.$$

Discrete GRA-step: If $J_{\text{foam}} > \epsilon$, the network does not continue blind descent. It initiates reductio ad absurdum, identifies nodes generating absurdity, and structurally revises logical constraints:

$$\Gamma^{(t+1)} = \text{Reductio} \left(\Gamma^{(t)}, \arg \max_{\Gamma} \varphi_{ij} \right).$$

This guarantees topological restructuring until the condition $\Delta^2 J > 0$ is satisfied.

8 In Silico Verification

8.1 Digital Twin of a Human

A digital twin D_H^{sim} with $N = 10^7$ nodes was created, modeling cognitive, emotional, somatic, social, and value levels. Typical contradictions were introduced:

- cognitive dissonance;
- emotional conflict;
- value mismatch;
- social role conflict;
- somatic anxiety.

8.2 Perturbation Test

Five hundred random perturbations were performed. For each, the following were measured:

- $J(t)$ — foam;
- τ — relaxation time;
- $\lambda_{\min}(\Delta^2 J)$ — minimum eigenvalue of the Hessian.

Results are presented in Table 1.

Metric	Baseline gradient method	GRA-NN
Final J	0.42 ± 0.11	0.000 ($\varepsilon = 10^{-6}$)
Relaxation time τ	870 steps	47 steps
$\lambda_{\min}$	-0.013 (saddle)	$+0.028$ (minimum)
Harmony index $HI = 1 - J/J_{\max}$	58%	99.8%

Table 1: Comparison of contradiction nullification methods in the digital twin

Conclusion: GRA-NN achieves complete nullification of contradictions in a finite domain and holds the state at a stable minimum.

8.3 Lyapunov Function

For the digital twin, a Lyapunov function was constructed:

$$V = J + \frac{1}{2} \|\nabla J\|^2.$$

It was shown that in all 500 trials $\frac{dV}{dt} < 0$, guaranteeing asymptotic stability of the zero state.

9 Practical Applications

1. **New psychotherapy.** The GRA-step formalizes cognitive therapy: instead of suppressing symptoms, it structurally eliminates the logical root of the contradiction.
2. **Personalized medicine.** A patient’s digital twin continuously nullifies somatic and emotional contradictions, selecting therapy consistent with values and physiology.
3. **Interpersonal conflict resolution.** A group of people is treated as a bounded domain D_G . GRA-nullification aligns participants’ goals and finds a stable state without contradictions.
4. **Education and learning.** Before introducing new knowledge, the student’s digital twin eliminates old cognitive foam, improving retention.
5. **Creative synergy.** Generative models with GRA-NN eliminate artifacts, creating semantically coherent works of art.

6. **Daily self-regulation.** A personal GRA assistant continuously measures J and suggests minimal structural changes to maintain harmony.

10 Conclusion

A specific human being is a computable bounded domain of knowledge. Within this domain, complete GRA-nullification of contradictions is achievable. The state of complete harmony is not a mystical abstraction but a stable manifold with zero foam and a positive-definite Hessian.

The formalism of GRA-NN with tensor-logical weights and the RAD algorithm shows that human initiative can be modeled as dynamic navigation along the Pareto front of goals. In silico verification confirms that after perturbations, the digital twin returns to $J = 0$, corresponding to a stable state of internal harmony.

Thus, “paradise” within the GRA paradigm is not an endpoint beyond life but a computably attainable state of complete elimination of internal contradictions in the bounded domain of a specific human.

References

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